

Chapter 2

THE SEVEN BASIC RULES OF HACKING: GENERAL ADVICE

Like boot camp or Candyland, this book is almost devoid of theory, but heavy on rules. Here are a few guidelines for keeping you healthy and happy:

Rule #1: Fear not!

Ignorance *is* bliss, anything worth doing is worth doing *wrong*, and two wrongs *can* make a right.

Rule #2: Don't take apart anything that plugs directly into the wall.

We will work almost exclusively with battery-powered circuitry. AC-powered things can kill you. AC adapters (wall-warts) may be used *only* after you have displayed proper understanding of the difference between insulation and electrocution.

Rule #3: It is easier to take something apart than put it back together.

Objects taken apart are unlikely to function normally after they are put back together, no matter how careful you are. Consider replacement cost before you open.

Rule #4: Make notes of what you are doing as you go along, not after.

Most wires look pretty much alike. As you take things apart make notes on which color wire goes to where on the circuit board, to what jack, etc. Especially important are the wires that go to the battery. Likewise, note what you add as you add it, what you change as you change it.

Rule #5: Avoid connecting the battery backwards.

This can damage a circuit.

Rule #6: Many hacks are like butterflies: beautiful but short-lived.

Many hacks you perform, especially early in your career, may destroy the circuit eventually. Accept this. If it sounds great, record it as soon as possible, and make note of what you've done to the circuit so you can try to recreate it later (see Rule #4).

Rule #7: In general, try to avoid short circuits.

Try to avoid making random connections between locations on a circuit board using plain wire or a screwdriver blade. This can damage a circuit—not *always*, but inevitably at the most inconvenient time.

Additional rules will emerge from time to time throughout the book, and are reiterated in Appendix D.